

Methodology

1. Data Preprocessing & Feature Engineering

The original dataset contained 3240 numerical features and 315 samples in the training set. To address the **curse of dimensionality** and reduce noise, following steps were performed:

- **Dropped non-informative columns:** "ID" was removed as it does not contribute to the prediction task.
- **Standardization:** All features were standardized to have zero mean and unit variance using `StandardScaler`.
- **Feature Selection:** We trained a **Random Forest classifier** on the full feature set and extracted the **top 50 most important features** based on the model's feature importance scores. This approach leverages tree-based models' ability to rank features effectively even in high-dimensional settings, ignoring the 'less contributing' features in the classification.

2. Model Architectures & Hyperparameters

We trained and evaluated three classification algorithms on the selected top 50 features:

- **Random Forest (for feature selection only)**
 - Number of trees: 1000
 - `random_state=42`
 - Was used to rank features by importance, and top 51 features used for the rest of the models.
- **Logistic Regression**
 - Max Iterations: 1000
 - No hyperparameter tuning was applied due to the small dataset size.
- **Support Vector Machine (SVM)**
 - Kernel: RBF
 - Regularization parameter $C = 50$
- **Ensemble(Logistic Regression + SVM)**
 - `LogisticRegression(C=30.0, L2 penalty, liblinear solve, max_iter=2000)`
 - `SVC(C=30, kernel=rbf, max_iter=5000)`
 - `VotingClassifier` used with 'soft' voting

All models were trained on the 51 selected features to maintain consistency and avoid overfitting due to high dimensionality.

3. Cross-Validation Scheme

Due to the limited number of samples (315 in training), we opted not to perform extensive hyperparameter tuning or cross-validation in this submission. `Test_set` was used only for testing, and `train_dataset` was only used to get the following metrics. Future work could involve 5-fold or stratified cross-validation for robust model selection.

4. Results Table (with respect to `test_set.csv`)

Model	Accuracy	AUCROC	Recall	Specificity	F1-score
Logistic Regression	0.6900	0.6732	0.4523	0.8620	0.55007
SVM	0.7200	0.7249	0.5476	0.8448	0.6216
Ensemble	0.7000	0.6900	0.3809	0.9310	0.5161

5. Discussion

Strengths

- Used a robust, interpretable feature selection technique (Random Forest) to reduce dimensionality from 3240 to 50 features.
- Applied two strong classification algorithms: logistic regression for baseline interpretability, and SVM for capturing non-linear decision boundaries. Further ensemble of both of these models was chosen to capture all nuances of the features in the dataset.

Limitations

- Small training size (315 samples) limited model generalization and hindered effective hyperparameter tuning.
- No cross-validation or ensemble strategies used, which could boost performance.
- Grid Searches and Searching optimal hyperparameters seemed too time and compute consuming, and thus were opted out of the choice.
- No deep learning architectures were used, which could have improved the solution.

Potential Improvements

- Use of **Stratified K-Fold Cross-Validation** for better generalization.
- Hyperparameter tuning with **GridSearchCV** or **Bayesian Optimization**.